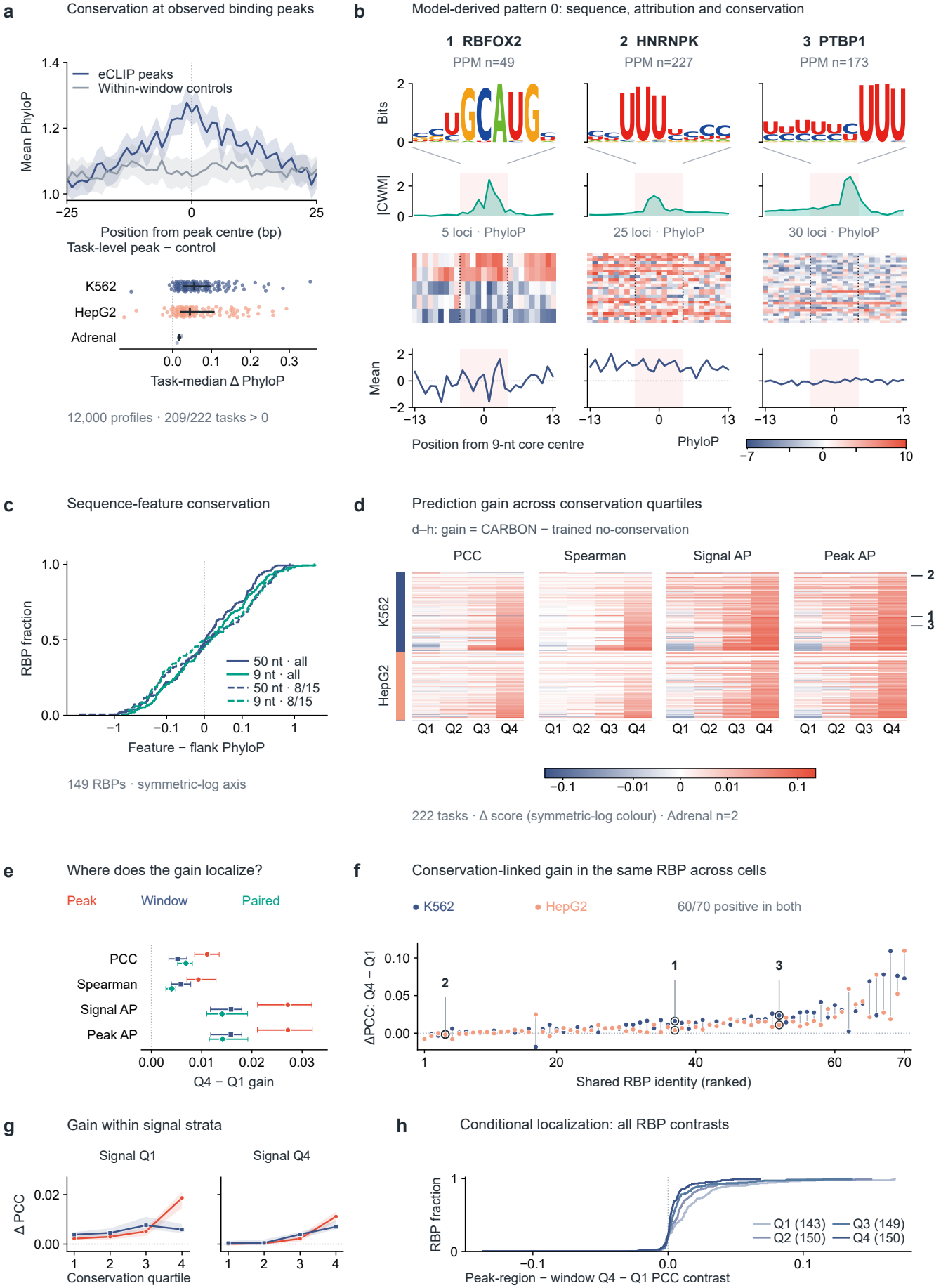

a
Conservation at observed binding peaks
eCLIP peaks
Within-window controls
Mean PhyloP
Position from peak centre (bp)
Task-level peak − control
K562
HepG2
Adrenal
Task-median Δ PhyloP
12,000 profiles · 209/222 tasks > 0
b
Model-derived pattern 0: sequence, attribution and conservation
1 RBFOX2
PPM n=49
2 HNRNPK
PPM n=227
3 PTBP1
PPM n=173
Bits
|CWM|
5 loci · PhyloP
25 loci · PhyloP
30 loci · PhyloP
Mean
Position from 9-nt core centre
PhyloP
c
Sequence-feature conservation
RBP fraction
50 nt · all
9 nt · all
50 nt · 8/15
9 nt · 8/15
Feature − flank PhyloP
149 RBPs · symmetric-log axis
d
Prediction gain across conservation quartiles
d–h: gain = CARBON − trained no-conservation
PCC
Spearman
Signal AP
Peak AP
K562
HepG2
Q1 Q2 Q3 Q4
222 tasks · Δ score (symmetric-log colour) · Adrenal n=2
e
Where does the gain localize?
Peak
Window
Paired
PCC
Spearman
Signal AP
Peak AP
Q4 − Q1 gain
f
Conservation-linked gain in the same RBP across cells
K562
HepG2
60/70 positive in both
ΔPCC: Q4 − Q1
Shared RBP identity (ranked)
g
Gain within signal strata
Signal Q1
Signal Q4
Δ PCC
Conservation quartile
h
Conditional localization: all RBP contrasts
RBP fraction
Q1 (143)
Q2 (150)
Q3 (149)
Q4 (150)
Peak-region − window Q4 − Q1 PCC contrast